



Fig. 2: New era of display

understanding and retention of subject matter.

Healthcare. ITAD technology facilitates advanced medical training and surgical simulations within healthcare settings. Surgeons can practise intricate procedures in a virtual environment, refining their skills without the need for cadavers or live patients. Additionally, ITAD-enabled medical imaging systems provide enhanced visualisation of diagnostic data, aiding in more accurate diagnoses and treatment planning.

Gaming and entertainment. The gaming industry has embraced ITAD technology to deliver unparalleled gaming experiences. By incorporating 3D touch interactions into gameplay, developers can create immersive virtual worlds where players can manipulate objects, navigate environments, and engage with characters in new ways. Furthermore, ITAD technology

enhances the viewing experience for multimedia content, such as movies and interactive storytelling experiences.

Design and engineering. ITAD technology facilitates collaborative prototyping and product visualisation in design and engineering fields. Designers and engineers can interact with virtual models in real time, making precise adjustments and annotations. This immersive approach to design accelerates the development process and fosters innovation by enabling more iterative and dynamic workflows.

Retail and marketing. ITAD technology has the potential to revolutionise the retail experience by enabling virtual try-ons for clothing and accessories. Customers can visualise how products look and fit in a three-dimensional space, enhancing their purchasing confidence and reducing the likelihood of returns. Additionally, immersive

advertising experiences powered by ITAD technology captivate audiences and drive engagement through interactive storytelling and product demonstrations.

Implications for human-computer interaction

The advent of ITAD technology signifies a paradigm shift in human-computer interaction, blurring the lines between the physical and digital realms. By providing tactile feedback and spatial awareness, ITAD interfaces bridge the gap between the virtual and tangible worlds, opening up new possibilities for communication and collaboration.

One of the key implications of ITAD technology is its potential to enhance accessibility for individuals with disabilities. By incorporating haptic feedback and intuitive gestures, ITAD interfaces cater to a broader range of users, including those with visual or motor impairments. Furthermore, the immersive nature of ITAD experiences can provide new avenues for therapy and rehabilitation, allowing individuals to engage in therapeutic activities within virtual environments.

Moreover, ITAD technology can redefine productivity and creativity in the workplace. By enabling more natural and intuitive interactions with digital tools, ITAD interfaces empower users to express their ideas more fluidly and collaborate more effectively. Whether brainstorming ideas in a virtual workspace or sketching designs in three dimensions, ITAD technology fosters innovation by removing barriers to creativity.

Challenges and future directions

While ITAD technology holds immense promise, it also presents